

MACHINE LEARNING FOR IOT
SRTA 3353
2025/2026
GROUP PROJECT (25%)

NODE-RED INTELLIGENT IOT SOLUTION USING MACHINE LEARNING

Part A : Project Question

In a team of maximum **3**, design and implement a practical Intelligent IoT solution using Node-RED as the integration platform and Machine Learning for decision-making.

Your system must demonstrate an end-to-end pipeline

1. **IoT Data Ingestion** (real sensors OR simulation OR dataset replay)
2. **Data Preprocessing** (cleaning, smoothing, feature creation)
3. **ML Model Inference** (classification/regression/anomaly detection)
4. **Actionable Output** (dashboard + alerts + recommendations)

You must build a solution for **ONE** IoT environment below and show how ML improves the system compared to a non-ML baseline (e.g., simple threshold/rules).

Link to Node-RED: <https://nodered.org>

Part B: IoT Environment

Choose **ONE** IoT Environment

A) Smart Transport / Mobility (high relevance)

Examples:

- Predict passenger crowd level (Low/Med/High) from time + count + weather
- Detect abnormal vibration/temperature patterns for vehicle maintenance
- Predict Wi-Fi service degradation risk (quality classification)

B) Smart Building / Campus

Examples:

- Predict occupancy from CO₂/temp/humidity
- Detect abnormal energy consumption (anomaly detection)
- Classify air quality risk (Good/Moderate/Unhealthy)

C) Smart Agriculture / Environment

Examples:

- Predict irrigation need (regression)
- Detect sensor failures (anomaly detection)
- Classify crop stress risk level

D) Industrial / Equipment Monitoring

Examples:

- Detect motor overheating anomalies
- Predict failure risk category (classification)
- Predict remaining useful life (simple regression approximation allowed)

Part C: Technical Requirements

1) Node-RED Flow (Core)

Your Node-RED flow must include:

- **Number of features** (at least 3 input features)
- **Input node** (MQTT / HTTP / Serial / file replay / simulated stream)
- **Preprocessing stage** (at least 2 preprocessing steps)
- **ML inference stage** (model output used in logic)
- **Dashboard** (at least 2 widgets: chart + gauge/status)
- **Action** (at least 1: alert OR recommendation OR automatic control)

2) Machine Learning Requirement (Core)

Use **at least ONE ML technique**:

- **Classification** (e.g., risk category)
- **Regression** (predict numeric)
- **Anomaly detection** (detect outliers/unusual behavior)

Minimum evaluation required:

- Classification: Accuracy + F1 (or Precision/Recall)
- Regression: MAE or RMSE
- Anomaly: explain evaluation method + show example detections (and if labels exist, precision/recall)

3) Baseline Comparison (Core)

Show a simple baseline approach (e.g., threshold rule) and explain why ML is better:

- Compare performance (metrics or qualitative evidence)
- Compare false alarms vs ML

Part D: Team Roles (Mandatory)

Each group must assign these roles **(1 per person)** and show evidence of contribution.

Student A — Data & IoT Pipeline Engineer

- Data source selection (sensor/sim/dataset)
- Node-RED ingestion nodes
- Data logging (CSV/DB/file)
- Data quality handling (missing/noisy)

Student B — ML Engineer

- Feature design + model selection
- Training + evaluation metrics
- Build inference method:
 - either Python REST microservice **OR**
 - built-in Node-RED logic (if permitted)
- Provide model explanation + limitations

Student C — System Integrator & Dashboard/Action Engineer

- Integrate inference into Node-RED decision logic
- Build dashboard (charts/gauges/indicators)
- Implement alerting/recommendation/control
- Demonstration scenario design (normal vs abnormal cases)

 **Important:** Marks include both team output and individual contribution.

Part D: Deliverables

Submit the required documents as the following **before 10th of May 2026**

1. **Node-RED flow export** (.json export text)
2. **Short report (5–8 pages)** including:
 - Problem statement + chosen IoT environment
 - System architecture diagram
 - Data description (features, sampling rate, volume)
 - Preprocessing steps
 - ML method + training process
 - Evaluation results + baseline comparison
 - Final decision logic + alerts/recommendations
3. **Demo evidence (3–6 minutes video OR screenshot sequence)** showing
 - Data streaming into Node-RED
 - ML inference output
 - Dashboard changing live/replayed
 - Alert/recommendation triggered
4. **Contribution log (1 page)**
 - Role assignment
 - What each student implemented
 - Git screenshots / version logs / Node-RED screenshots (anything verifiable)

Part E: Academic Integrity Note

1. Each group must submit unique flow/model.
2. If using public datasets, cite them and explain feature work.
3. Any copied flow without modification receives low/no marks.
4. Similarity Index and AI Similarity Index must be below 20%.

References

1. Hunkeler, U., Truong, H. L., & Stanford-Clark, A. (2008). *MQTT-S—A publish/subscribe protocol for Wireless Sensor Networks*. Proceedings of the 3rd International Conference on Communication Systems Software and Middleware (COMSWARE). [\[ieeexplore.ieee.org\]](https://ieeexplore.ieee.org)
2. “Open Source IoT-Based SCADA System for Remote Oil Facilities Using Node-RED and Arduino Microcontrollers.” (2020). IEEE Xplore conference/publication. [\[ieeexplore.ieee.org\]](https://ieeexplore.ieee.org)
3. “Real-time Feedback in Node-RED for IoT Development: An Empirical Study.” (2020). IEEE Xplore. [\[ieeexplore.ieee.org\]](https://ieeexplore.ieee.org)
4. “Node-RED Workflow Manager for Edge Service Orchestration.” (2022). IEEE Xplore. [\[ieeexplore.ieee.org\]](https://ieeexplore.ieee.org)
5. “Smart Home Automation with MQTT Using ESP32 and Node-RED.” (2025). IEEE Xplore. [\[ieeexplore.ieee.org\]](https://ieeexplore.ieee.org)
6. “Real-Time Monitoring of DC Microgrids of Solar Farms Using an MQTT-based IIoT Device integrated with Node-RED.” (2025). IEEE Xplore. [\[ieeexplore.ieee.org\]](https://ieeexplore.ieee.org)

Assessment Rubrics

Criterion	Excellent (Full marks)	Good	Satisfactory	Poor / Incomplete	Marks
A. Real-World Problem & IoT Context	Problem is clearly defined, realistic, and strongly aligned with a real IoT environment; constraints and objectives are well justified	Problem is relevant and realistic but with minor gaps in justification or context	Problem is stated but weakly linked to real-world IoT use	Problem unclear, unrealistic, or poorly justified	4
B. Node-RED Data Pipeline & Architecture	Complete end-to-end pipeline with correct data ingestion, preprocessing, logical flow structure, and robustness (handles noise/missing data)	Pipeline works correctly with minor design or robustness issues	Basic pipeline works but lacks preprocessing depth or good structure	Incomplete, incorrect, or poorly functioning pipeline	6
C. Machine Learning Method & Evaluation	ML technique is well-chosen and justified; features are appropriate; training and evaluation are correct; metrics clearly reported; baseline comparison included	ML used correctly with evaluation, but justification or baseline comparison is limited	ML used but weak feature design or incomplete evaluation	ML missing, incorrect, or no proper evaluation	8
D. Integration, Dashboard & Actionable Output	ML inference is fully integrated into Node-RED logic; dashboard is clear and meaningful; alerts/recommendations are accurate and useful	Integration works with functional dashboard and basic actions	Limited integration; dashboard or actions are minimal	No meaningful integration, dashboard, or action	5
E. Documentation & Individual Contribution Evidence	Clear report, architecture diagram, demo evidence provided; individual roles and contributions clearly documented	Documentation mostly clear with minor gaps in clarity or contribution evidence	Basic documentation provided but weak contribution clarity	Poor or missing documentation and contribution evidence	2
Total					25 marks